

EXP NO: 2	SUPPORT VECTOR MACHINE (SVM) AND RANDOM FOREST FOR BINARY & MULTICLASS CLASSIFICATION
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AIM

To build classification models using **Support Vector Machines (SVM)** and **Random Forest**, apply them to a dataset, and evaluate the models using performance metrics like accuracy and confusion matrix.

CODE1:

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3, random_state=42)

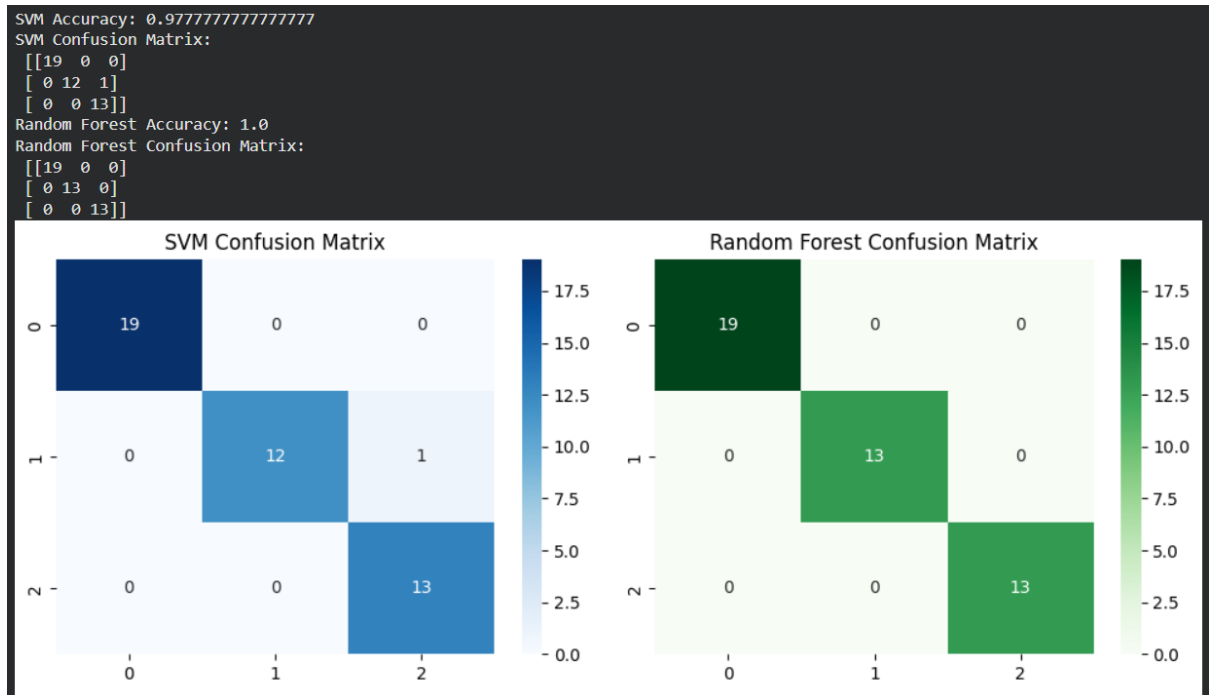
# Part A: SVM
svm_model = SVC(kernel='linear')
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:\n", confusion_matrix(y_test, y_pred_svm))
```

```
# Part B: Random Forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix:\n", confusion_matrix(y_test, y_pred_rf))

# Visual comparison
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
sns.heatmap(confusion_matrix(y_test, y_pred_svm), annot=True, cmap='Blues', fmt='d')
plt.title("SVM Confusion Matrix")

plt.subplot(1, 2, 2)
sns.heatmap(confusion_matrix(y_test, y_pred_rf), annot=True, cmap='Greens', fmt='d')
plt.title("Random Forest Confusion Matrix")
plt.tight_layout()
plt.show()
```

OUTPUT1:



CODE2:

```

import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Wine dataset
wine = load_wine()
X = wine.data
y = wine.target

# Scale Features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3, random_state=101, stratify=y)

# Model Training
svm_wine = SVC(kernel='rbf', C=1.0, gamma='scale')
svm_wine.fit(X_train, y_train)

rf_wine = RandomForestClassifier(n_estimators=100, max_depth=5, random_state=101)
rf_wine.fit(X_train, y_train)

y_pred_svm = svm_wine.predict(X_test)
y_pred_rf = rf_wine.predict(X_test)

print("SVM Accuracy Score:", accuracy_score(y_test, y_pred_svm))
print("Random Forest Accuracy Score:", accuracy_score(y_test, y_pred_rf))

# Dual Plot Confusion Matrices (Single-Line Parameters)
cm_svm = confusion_matrix(y_test, y_pred_svm)
cm_rf = confusion_matrix(y_test, y_pred_rf)
names = wine.target_names

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

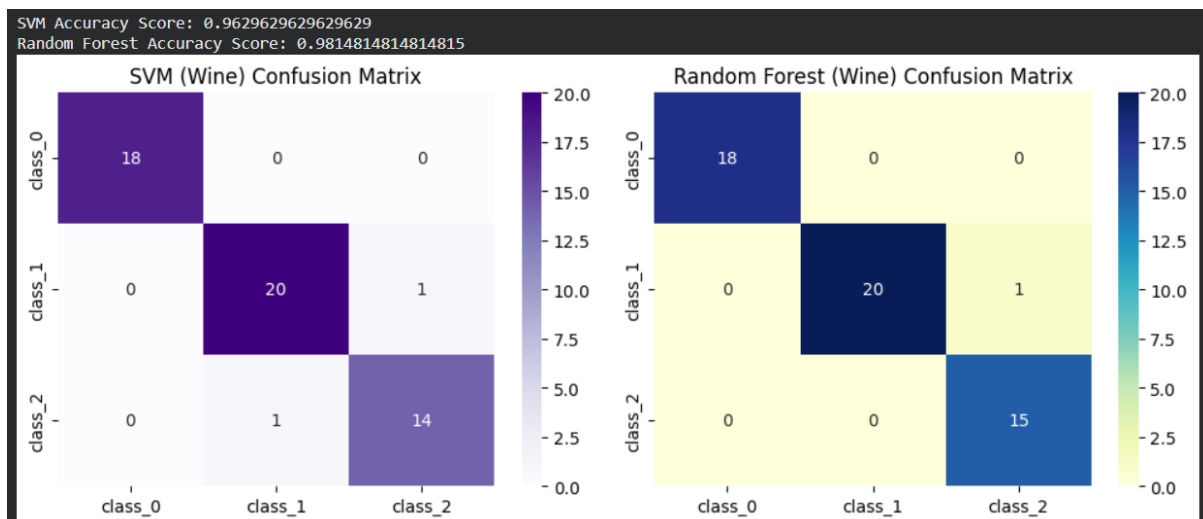
sns.heatmap(cm_svm, annot=True, cmap='Purples', fmt='d', xticklabels=names, yticklabels=names, ax=axes[0])
axes[0].set_title("SVM (Wine) Confusion Matrix")

sns.heatmap(cm_rf, annot=True, cmap='YlGnBu', fmt='d', xticklabels=names, yticklabels=names, ax=axes[1])
axes[1].set_title("Random Forest (Wine) Confusion Matrix")

plt.tight_layout()
plt.show()

```

Output 2:



RESULT:

The Support Vector Machine (SVM) and Random Forest algorithms were successfully implemented for both binary and multiclass classification tasks. The models were trained and tested on the given dataset, and both achieved good accuracy.